Exact Segmentation Invariance and a Negative Feasibility Result for Tokenization-Oriented Machine Translation Policy Gradients

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Abstract

We investigate whether changing target-side segmentation can provide a clean intervention for reinforcement learning in machine translation. The project produced two complementary negative results. First, for a policy that samples a surface string and then samples a conditionally normalized segmentation, ordinary response-level policy gradients with string-only reward are exactly invariant to the segmentation sampler. Token-normalized or position-weighted updates can break this invariance, but then the effect belongs to the estimator. Second, a normalized piece-length tilt in NLLB-200 distilled 600M changed the complete autoregressive policy rather than merely resegmenting fixed strings. On a frozen English–German calibration set it increased token fertility while reducing both the fixed string reward and an independent translation metric. The prespecified launch screen was therefore not passed, and no research training run was launched. Together, these results clarify why exact behavior likelihood is necessary but insufficient for identifying a useful tokenization-only intervention in sequence-level reinforcement learning.

1 Motivation

Subword regularization and BPE-dropout show that sampling alternative segmentations can improve supervised machine translation (Kudo, 2018; Provilkov et al., 2020). This does not immediately imply a useful reinforcement-learning intervention. Policy-gradient training requires the likelihood of the process that generated an action, and changing the segmentation may either leave a string-level estimator unchanged or alter the represented autoregressive policy itself. Exploration and reward scaling are already known to matter in neural machine translation policy gradients (Kiegeland and Kreutzer, 2021).

We report the completed tokenization branch of a larger project. Our scope is deliberately narrow: estimator validity and intervention feasibility. We do not report an effect of tokenization on reinforcement-learning outcomes because the intervention failed its calibration gate before training.

2 Same-string intervention

Let a string policy sample a surface string s, after which a segmentation sampler draws a path z:

$$P_{\theta,\phi}(s, z \mid x) = P_\theta(s \mid x) q_\phi(z \mid s, x). \quad (1)$$

For string-only reward $R(s)$ and a normalized conditional sampler, define the two expected score-function gradients as

$$g_\theta = \mathbb{E}_{s,z}[R(s) \nabla_\theta \log P_{\theta,\phi}(s, z \mid x)] \quad (2)$$
$$= \mathbb{E}_s[R(s) \nabla_\theta \log P_\theta(s \mid x)], \quad (3)$$
$$g_\phi = \mathbb{E}_{z|s}[\nabla_\phi \log q_\phi(z \mid s, x)] = \mathbf{0}. \quad (4)$$

Thus a separately sampled segmentation cannot change the expected ordinary string-policy gradient. Scoring the alternate token path with the base model's raw autoregressive likelihood would instead score a process that did not generate the sample.

2.1 Exact finite verification

We enumerate a categorical policy over 3 strings and 5 joint string–path outcomes. Some strings admit short and long paths; reward depends only on the string. We enumerate ordered batches of size 2 and compare exact expectations with 200,000 Monte Carlo batches.

Table 1 varies the probability of the short path. The response-mean string gradient changes by at most 5.55e-17, and the segmentation-parameter gradient remains zero. Dividing by the random total number of tokens in each batch instead changes expected sample weighting and yields a nonzero segmentation gradient.

Estimator	$q = 0.25$	$q = 0.75$
Response mean, ϕ gradient	0.0	0.0
Global token mean, ϕ gradient	0.01198	0.01906

Table 1: Exact segmentation-sampler gradients as the short-path probability q varies. The global token mean has a sampled, path-dependent denominator.

Tilt minus control	Mean	Lower	Upper
Tokens per word	+0.031	+0.020	+0.043
chrF++ reward	-0.021	-0.032	-0.010
Corpus BLEU points	-2.20	-3.84	-0.62
Actions incl. EOS	+0.146	+0.068	+0.225
Decoded words	+0.010	-0.043	+0.063

Table 2: Paired source-bootstrap intervals on the frozen calibration set. The chrF++ reward is on its normalized scale; BLEU is reported in points.

The maximum Monte Carlo error for the response-mean string-gradient coordinates is 2.04e-04; its segmentation-gradient error is 1.56e-04. Corresponding errors for the global-token estimator are 4.28e-05 and 2.13e-06. These checks validate the implementation rather than provide translation-quality evidence.

3 Autoregressive length tilt

Because the same-string construction leaves no ordinary-gradient mechanism, we also tested a broader decoder-policy intervention in NLLB-200 distilled 600M, a released NLLB model (NLLB Team et al., 2024). At each decoding step, a fixed bias favored shorter visible SentencePiece tokens and the categorical policy was renormalized. The sampled token then changed the next decoder input, position, attention state, future distribution, and stopping time. This is a total autoregressive policy intervention, not reassignment of a segmentation to an otherwise fixed translation.

The frozen comparison used a tilt coefficient of 0.20 and temperature 0.70. The control used the entropy-matched temperature 0.70. Entropy was measured on shared teacher-forced prefixes; the residual was -0.0099 nats against a tolerance of 0.05 nats. The calibration contained 975 unique Tatoeba sources and 2 paired rollout replicates, giving 1,950 generated outputs. Human scores were not used.

Table 2 shows the central feasibility result. Fertility increased, but the reward and independent metric both decreased. The action count changed while decoded word count was nearly stable, supporting a tokenization shift, but not a competence-preserving one. All 1,950 outputs ended in EOS; truncation, empty-output, and non-EOS special-token counts were respectively 0, 0, and 0. The prespecified quality screen remained inconclusive about whether both losses were within acceptable limits after its sole bounded extension. The pilot was not launched.

4 Discussion

The two studies rule out different shortcuts. Exact joint likelihood does not make a same-string segmentation sampler causally active under an ordinary string-reward estimator. Conversely, passing alternate token identities through an autoregressive decoder supplies a mechanism, but it also changes the policy's initial competence and surface-string distribution. A fertility change under token-normalized credit assignment is an estimator effect; a fertility change under the length tilt is a total decoder-policy effect.

This distinction also bounds the negative result. We do not conclude that tokenization is irrelevant to reinforcement learning. We conclude that the tested clean construction is invariant, while the tested non-invariant construction did not satisfy its launch gate.

5 Conclusion

The tokenization branch is complete as a validation and feasibility study. It provides an exact invariance result, tests where alternative reductions break that invariance, and records a negative real-model calibration. It does not establish a translation-learning effect, because training was correctly stopped before that estimand was evaluated.

Limitations

The exact harness is intentionally finite and does not model neural decoder states. The real-model screen covers one model family, one language direction, one calibration split, and one fixed tilt dose. Entropy matching does not make two policies equivalent. The calibration estimates initial behavior only and cannot determine whether later reinforcement learning would recover from the quality loss.

Ethical Considerations

The real-model study used the documented non-commercial research terms for the NLLB checkpoint and attributed Tatoeba data. No human judgments were newly collected. No model weights were released or trained after the feasibility screen failed.

References

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